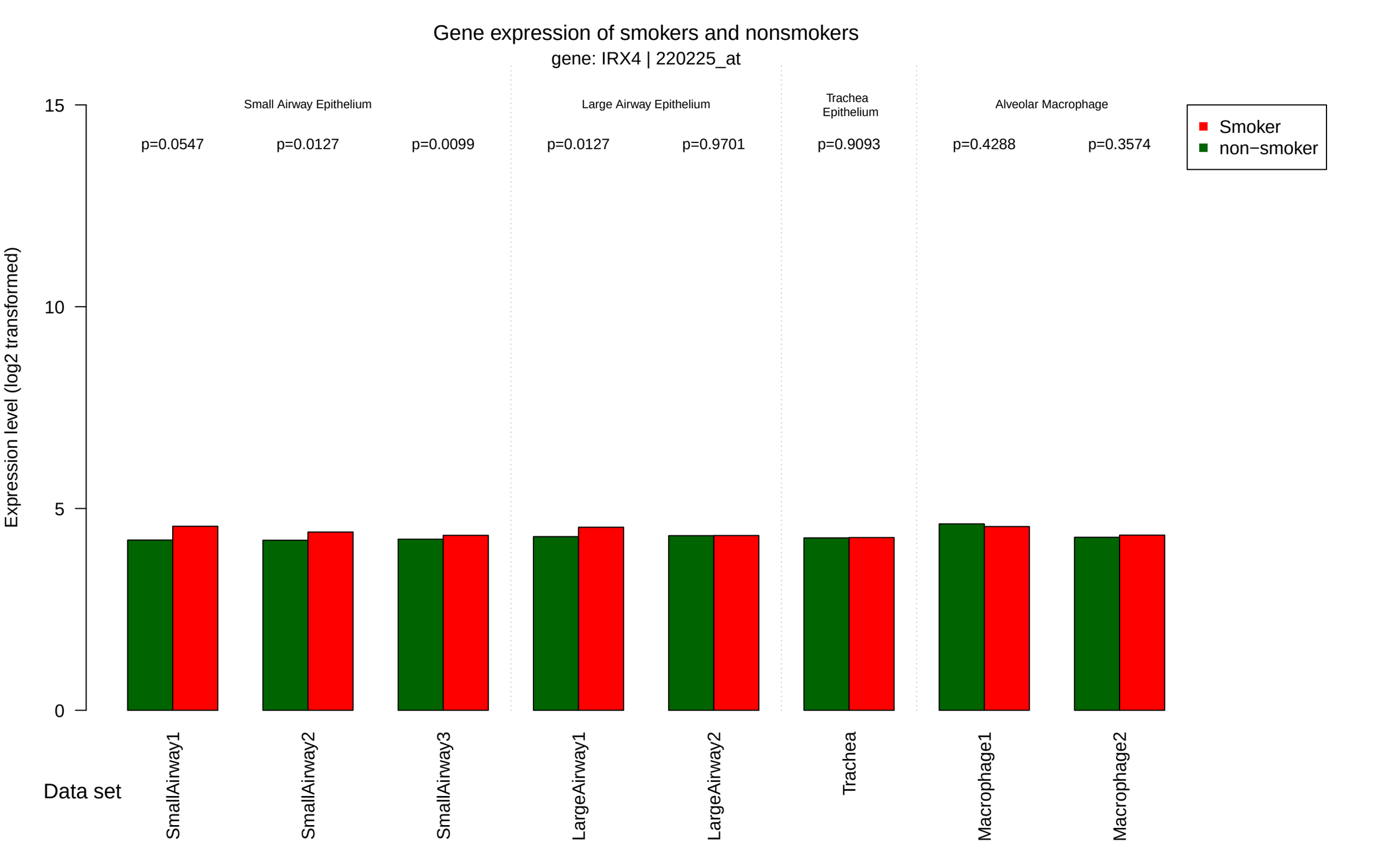
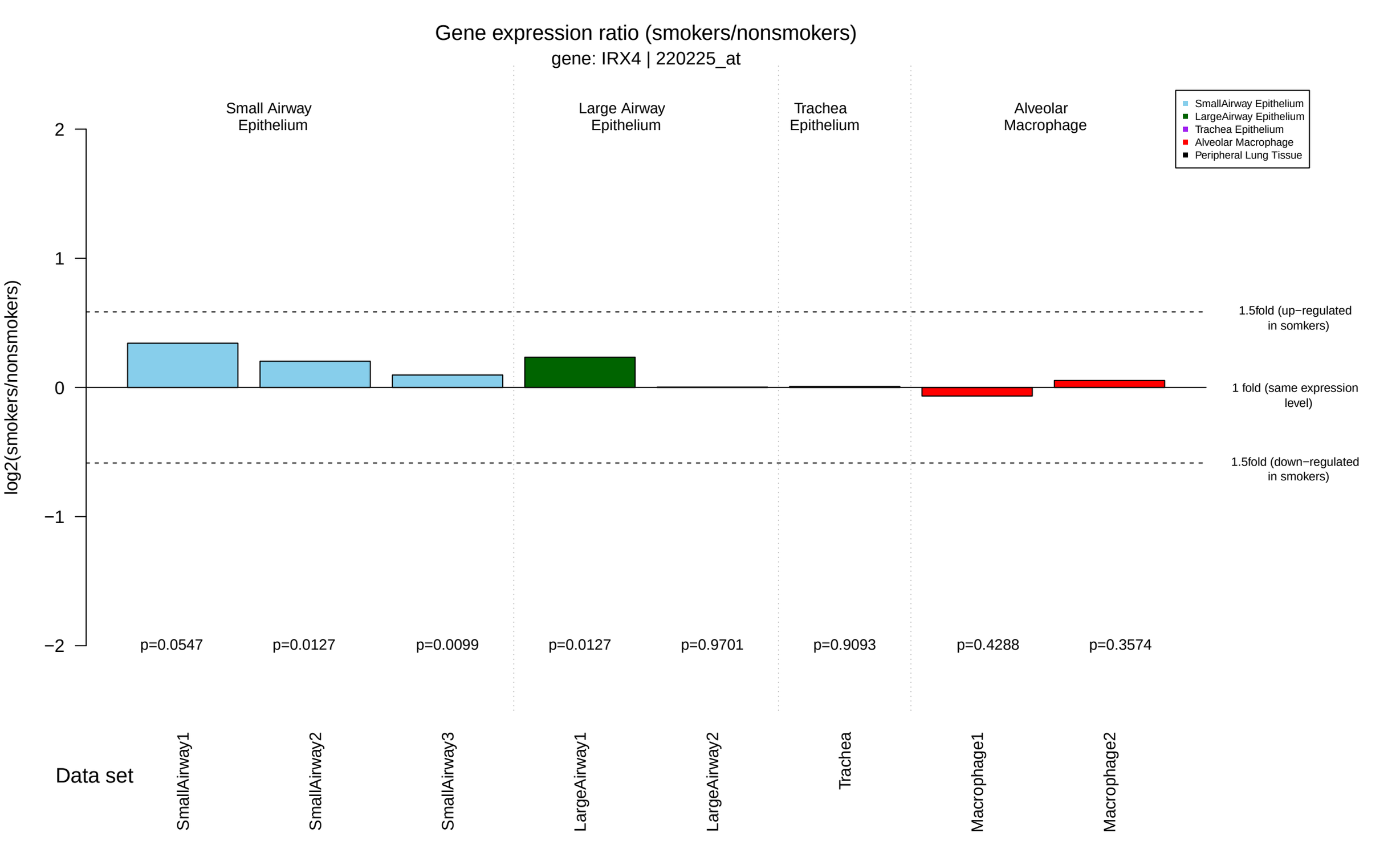
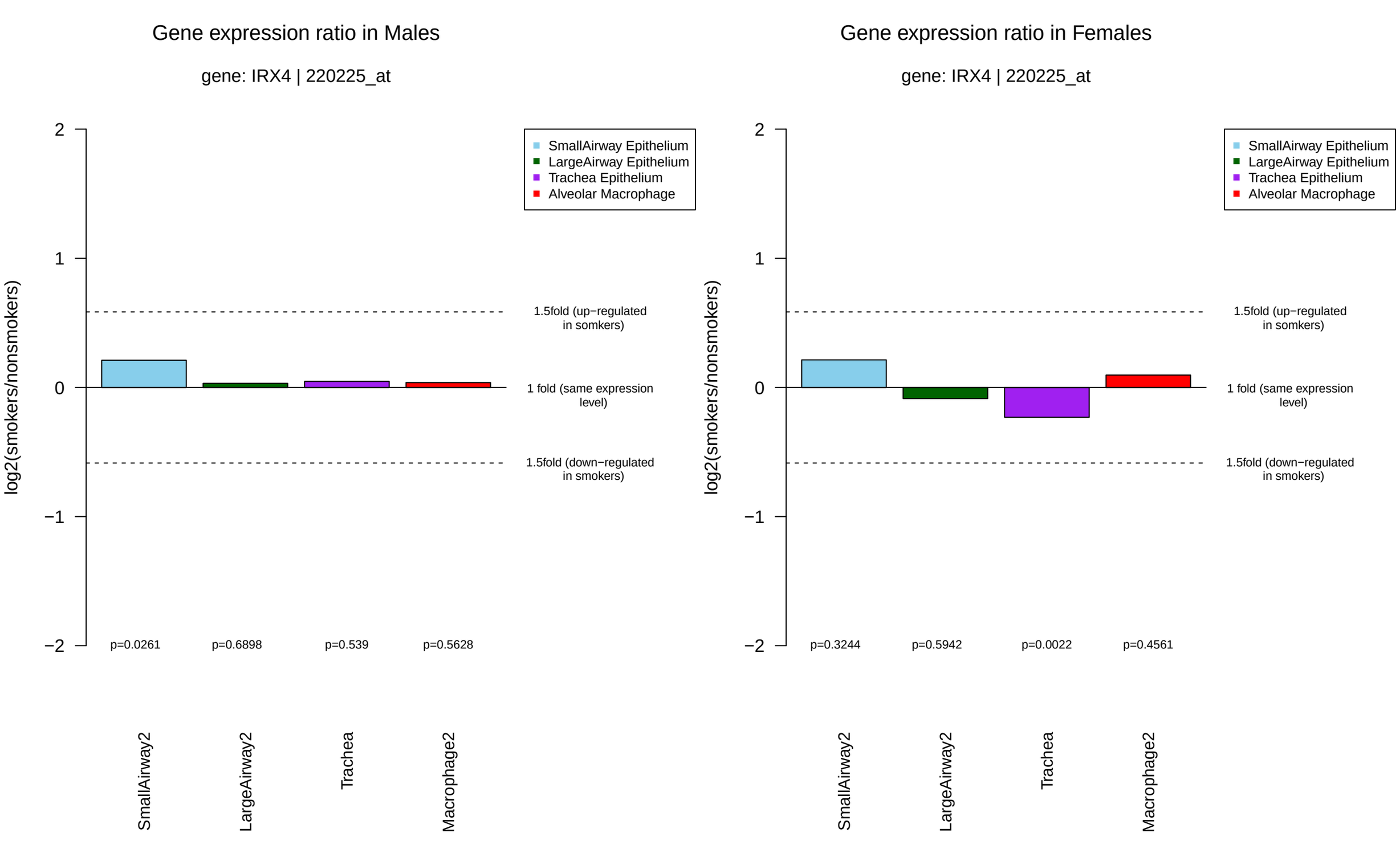
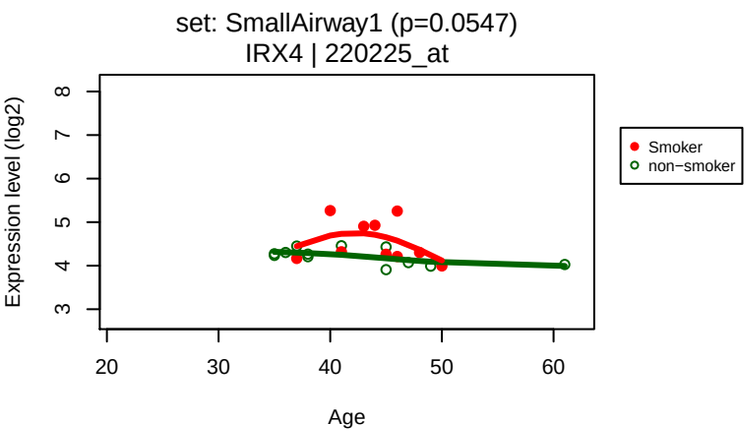
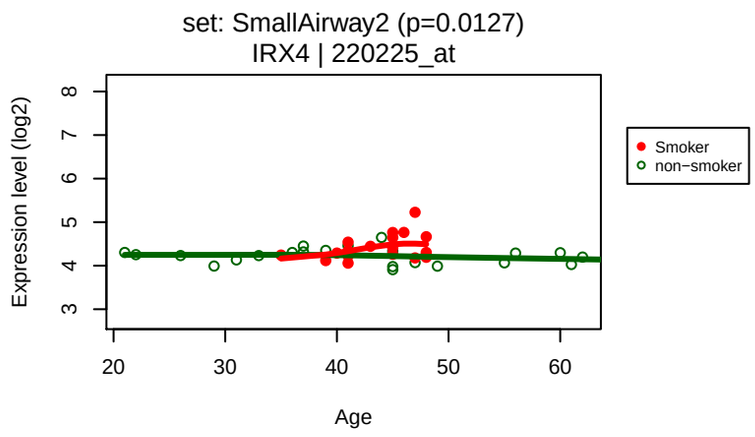
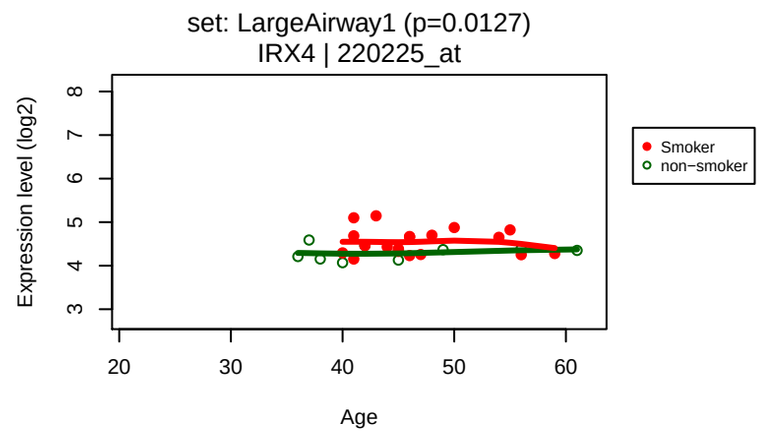

Gene expression of smokers and nonsmokers
gene: IRX4 | 220225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0547
p=0.0127
p=0.0099
p=0.0127
p=0.9701
p=0.9093
p=0.4288
p=0.3574
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

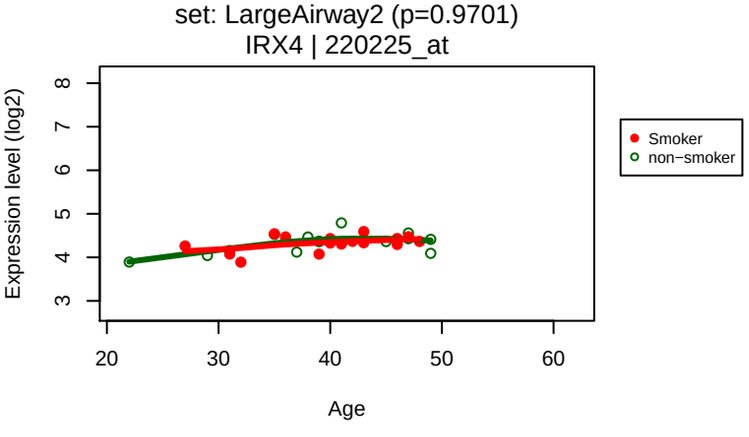

Gene expression ratio (smokers/nonsmokers)
gene: IRX4 | 220225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0547
p=0.0127
p=0.0099
p=0.0127
p=0.9701
p=0.9093
p=0.4288
p=0.3574
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

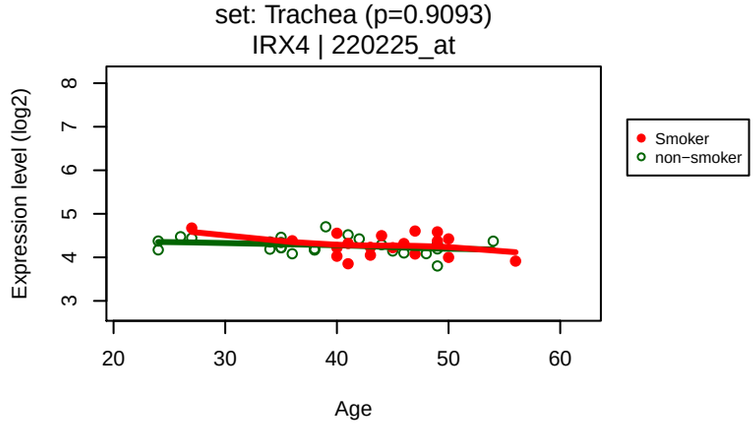

Gene expression ratio in Males
gene: IRX4 | 220225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0261
p=0.6898
p=0.539
p=0.5628
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRX4 | 220225_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3244
p=0.5942
p=0.0022
p=0.4561
SmallAirway2
LargeAirway2
Trachea
Macrophage2

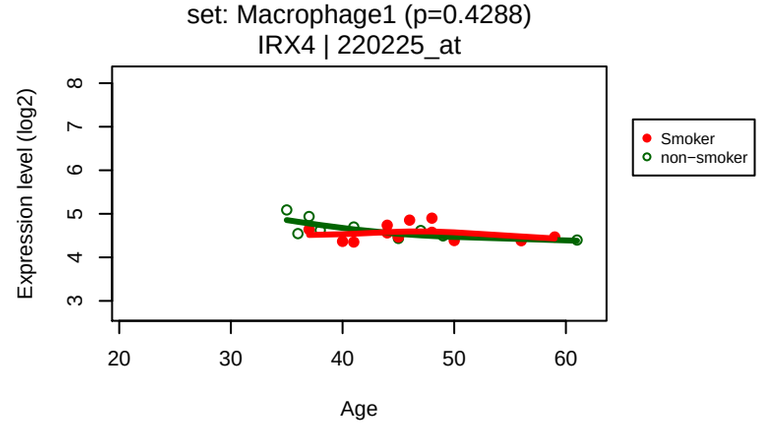

set: SmallAirway1 (p=0.0547)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker

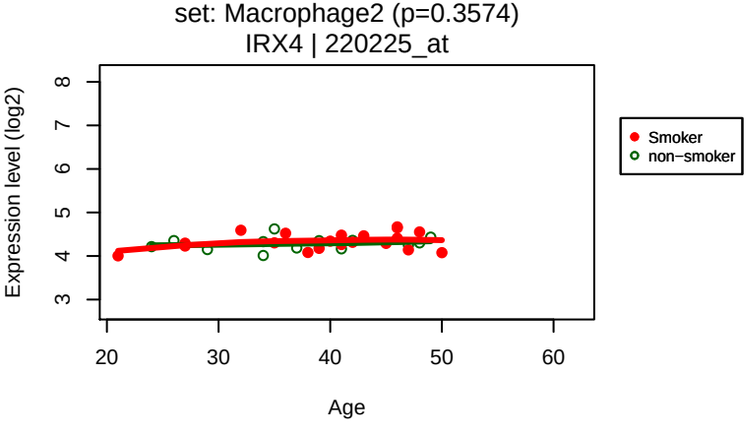

set: SmallAirway2 (p=0.0127)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker

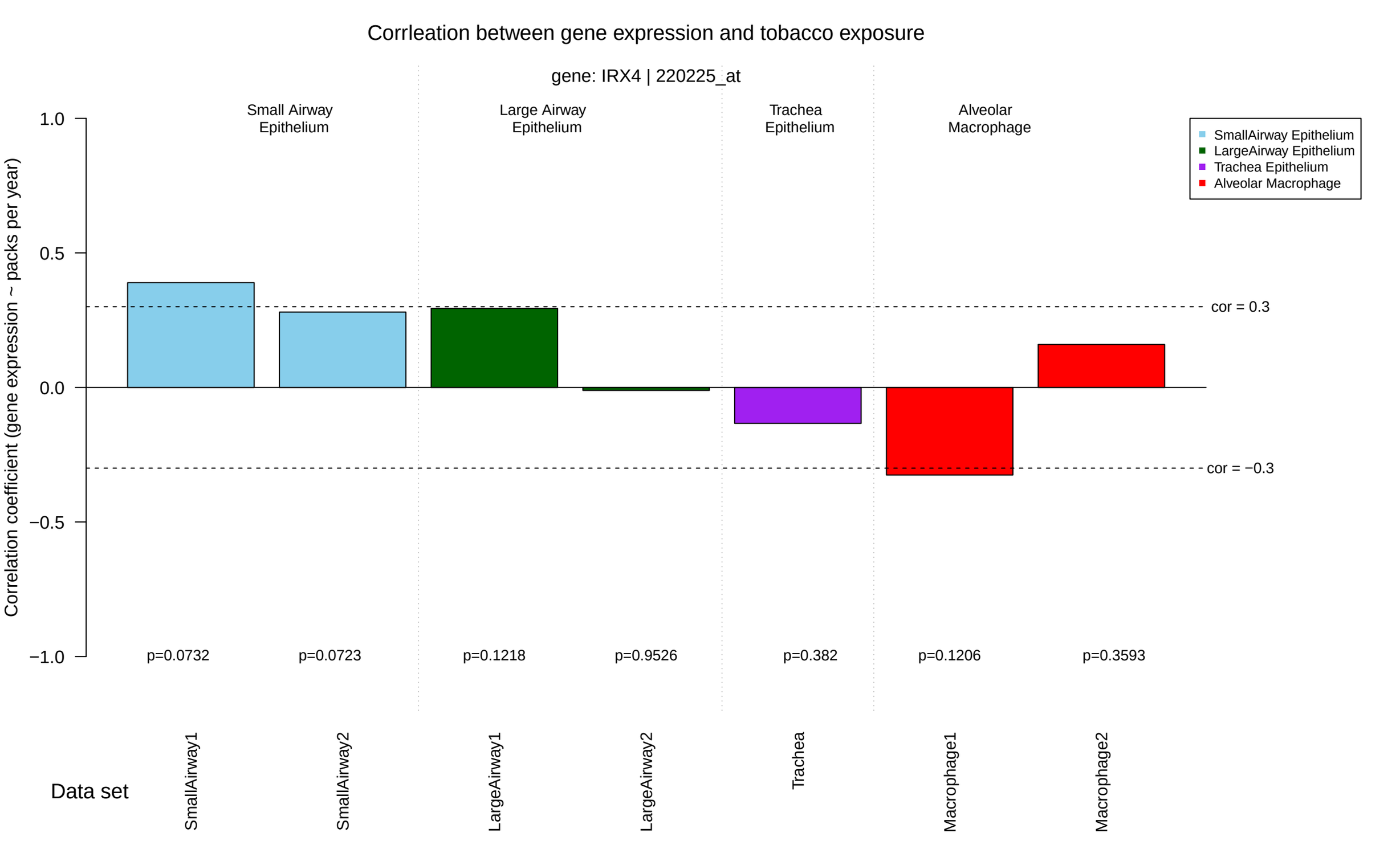

set: LargeAirway1 (p=0.0127)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9701)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9093)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4288)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3574)
IRX4 | 220225_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRX4 | 220225_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0732
p=0.0723
p=0.1218
p=0.9526
p=0.382
p=0.1206
p=0.3593
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2